

Overview

The paper extends the Assembly Calculus (AC) framework by replacing k-winners-take-all with E%-winners-take-all, a selection rule derived from gamma-cycle dynamics in which the number of firing neurons is determined by the most strongly driven neuron rather than fixed in advance. A second modification assigns a fraction of synapses negative weight, derived from cortical excitation-inhibition balance. The architecture consists of a stimulus area (S) projecting into a memory area (M), where neurons form recurrent connections among themselves and are subject to multiplicative Hebbian plasticity. Projections between multiple memory areas are described but not simulated. A fixed subset of S neurons is clamped active, and the resulting firing pattern in M is followed until it reaches a fixed point. The authors propose stricter formation criteria than AC: the active set must be stationary across iterations, no new neurons may join it, and its synaptic density must exceed that of the memory area. Under these criteria they report emergent assembly sizes controlled by inhibition, faster convergence than AC at most learning rates, near-complete recovery of an assembly when its stimulus is re-presented, and reduced overlap between assemblies stored in the same area.

Major comments

The retrieval experiment re-presents the identical stimulus to an unchanged network, so it asks whether a fixed point is a fixed point - a property the formation criterion already guarantees by construction. Medians of exactly 1.00 with zero-width interquartile ranges are the expected outcome, not evidence of quality. The same reasoning explains why the AC baseline scores lower: its stopping rule never required a fixed point, as the authors themselves demonstrate in Fig 2b. The comparison therefore measures a difference in definitions.

The metrics as constructed also fail to penalize degenerate solutions. A model that allocated a single neuron per stimulus would achieve perfect retrieval and zero overlap while storing nothing. That this is a live failure mode is evident from the authors' need to impose $|A|_{\min} = 6$ by hand.

Further, the claims about "efficient storage" and "efficient representation of information" are not supported. Nothing in this model is optimized, due to the lack of error signal - the system is not forced to learn optimally. Hence the problem of new ensembles destroying old ones that authors report.

The question about introducing an error-signal is a wide one, since it also requires changes to the original architecture (and, frankly, it can be debated from a biological plausibility standpoint, although I personally would try it to see what changes). However, a simple approach may suffice.

To separate real attractor stability from a tautology of the formation criterion, retrieval should be tested with a degraded cue, not the identical stimulus: present a random subset of the original S neurons (varying the retained fraction) or add noise/rewiring to the synapses post-formation, and plot $|A|_{\text{rec}}/|A|$ against degradation level. A real basin of attraction degrades steadily; a model that just re-finds its own fixed point should collapse once the cue departs from training. To rule out the degenerate single-neuron-per-stimulus code, the

authors should report retrieval fidelity and pairwise overlap as a function of the number of assemblies stored in one area ($n = 1 \dots N$), alongside total neurons committed. A trivial code would show near-zero overlap and perfect retrieval independent of n , with committed neurons scaling minimally with n ; departure from that signature, and the point where retrieval starts to fail, would give a real measure of capacity and directly quantify the forgetting effect already visible in Fig 4c.

Minor comments

1. Survival bias

Reported medians (size, density, formation time, recovery) are computed only over runs meeting the formation criteria, and success rate itself varies with β and ω_{inh} (Fig 2d). This means that parameters reported are potentially biased by varying β and ω_{inh} .

2. Model comparisons vary multiple parameters at once.

E%-WTA and AC differ in p_s (0.5 vs 0.1), k_s (200 vs 37), and the presence of inhibitory synapses simultaneously. Differences in convergence speed or retrieval cannot be attributed to the selection mechanism alone without a separate test isolating each change.

3. Parameters were tuned on the same data used to report results.

ω_{inh} and β were selected by grid search to minimize failure rate (Fig 2d), then reused for all headline comparisons (Table 2, Figs 3–4) with no held-out validation that this choice generalizes.

Conclusion

I believe the work is interesting at its core and explores a complicated subject of unsupervised memory formation. Whilst the ‘unsupervisedness’ of the real brain’s memory formation process is a subject to debate (one may argue that error signal comes with an emotional response), the endeavor is great.